

Assignment 3 — Difference-in-Differences

COVID-19 Lockdowns and Interest in Sourdough Bread

Choose your method. For this assignment you complete **one** of three options — matching, instrumental variables, or difference-in-differences — whichever best fits the research you care about. All three share the structure below; this file is the **Difference-in-Differences** option. Submit one knitted R Markdown file and include the AI-use statement. **Four blocks:** A and D are your own thinking (most weight), B is AI-assisted execution, C tests whether you catch AI's mistakes. Late penalty 10%/day.

Video (★). The ★ part below requires a video of no more than 3 minutes explaining how you got your answer and what it means.

The data. Use `a4_p1_sourdough_trends.csv` (on the course site): Google Trends popularity (*hits*) of several search terms over time, including “sourdough” (in *keyword*). **Context (adapted from Huntington-Klein & Nencka):** early in the pandemic, stores ran out of yeast and homemade sourdough surged. We ask whether COVID lockdowns actually increased interest in sourdough. Treat **March 15, 2020** as the “start of the pandemic.”

Research question. Did the onset of COVID-19 lockdowns causally increase interest in sourdough bread?

Block A — Setup & Judgment (no code)

Choose a journal you like where a study like this could end up (clinical, econ, sociology, up to you). Using that journal's tone and structure, write the introduction to an interesting paper answering this question. You can limit yourself to 1-2 pages max here (no need to fill all that space either). Some possible questions you could address include:

- Why should anyone care?** In one paragraph, motivate this question from a physician's, a researcher's, and a policy-maker's vantage point. What would each do differently if they knew the answer?
- Anticipate the selection.** What is the underlying comparison in an ideal world, and what do you have in the data? Which confounders can you observe in the data, and which can you not? What are the main identification challenges?
- State the identifying assumption in one sentence.** Complete precisely: “A regression of *spending on HDHP enrollment* recovers the causal effect only if ...” Then give one reason to believe it and one reason to doubt it *in this setting*.

Block B — AI-Assisted Execution (use AI deliberately)

Use **AI to write the code** — fluent, well-directed use is the goal. For **one** part, paste in the prompt you gave, the raw code returned, and a sentence on what you corrected or redirected (your prompt + critique artifact).

- Clean and choose your keywords.** Select the control keyword(s) you will use, choose measurement/transformations, and handle missing data. Describe and justify your choices.

- b. **Visualize.** Plot *hits* over time with a separate trend per keyword, and a vertical line at March 15, 2020. What does the picture suggest, and does it support a particular hypothesis?
- c. **Set up the design.** Choose and justify your control group (statistically if possible). Write your estimating equation as you would in a published paper, and discuss the optimal standard-error calculation.
- d. **Estimate.** Estimate the difference-in-differences model for both a static and dynamic treatment effect using an estimator of your choice. Report and interpret your results. What do you conclude about the effect of the pandemic on sourdough interest?

Block C — The Error Hunt (catch what AI gets wrong)

A student asked an AI assistant to run this analysis and pasted the reply below without checking it. **The code runs without any error message** and the numbers look plausible — but it has **at least three consequential problems** (a mix of coding bugs and interpretation mistakes).

Reproduce it on the real data, then find and fix the problems. For each: (i) what is wrong, (ii) the corrected code or statement, (iii) one sentence on how it changes the conclusion. Flag any extra problem you find that we did not plant.

The AI-generated analysis:

```
library(tidyverse); library(fixest)
df <- read_csv("a4_p1_sourdough_trends.csv")

df <- df %>%
  mutate(post      = date >= as.Date("2020-03-15"),
         treated = keyword == "sourdough")

# Difference-in-differences
did <- feols(hits ~ post + treated + post:treated, data = df)
summary(did)                                # default standard errors

# "The coefficient on post:treated (about 25) is the level of sourdough
# popularity after the pandemic began."
```

Block D — Interpret & Defend

- a. ★ **Translate for a decision-maker.** A trends analyst asks: “How much did the pandemic actually boost sourdough interest, above and beyond what was happening anyway?” Answer in plain language in your video. Be explicit about what the diff-in-diff nets out and what could still be confounding it (e.g., other things that changed in March 2020).
- b. **Put your results into context.** Assume that you fully believe your results. Write the first two paragraphs of a Discussion section that interprets these results in a way that you think an editor would find compelling. You can compare them to external literature, report clinical/economic significance, or find some other “headline” way to showcase these results.
- c. **Revisit your assumption.** Return to your one-sentence identifying assumption (parallel trends). After seeing the data, are you more or less worried about it? Create and report at least one

piece of evidence — ideally a pre-trends comparison — that speaks to its credibility. Why is your evidence imperfect?

Mini Referee Report

Peer review is the part of the job no AI can do for you. AI assistants summarize happily but rarely land the concern that actually threatens a result — and they tend to agree with whatever they are reviewing. That judgment is yours.

Assigned paper: You may choose one paper from three options posted in our GitHub folder under HAD5744\Lecture11_DifferenceinDifferences2\RefereeReportOptions.

Read the assigned paper and write a short referee report (about one page, 400–600 words) naming the **two or three MAJOR concerns** — the ones that would change your confidence in the main result — not minor issues of writing. Structure it as: (1) a 2–3 sentence summary of the claim and how it is identified; (2) two or three major concerns, each with why it threatens the central result and what evidence would resolve it; and (3) a verdict naming the single most important revision.

Using AI here. You may use AI to understand the paper, but the concerns must be your own — its first list is usually generic and agreeable. If you used it, note in one line what it suggested and whether you agreed.